

Sales Analysis Report

Project: retail-analytics-eda

This report presents the findings of the sales analysis phase. The analysis was performed in SQL (SQL Server / SSMS) on two views `seasonality_VS_margin` and `sales_inventory` and visualised in Power BI. Where the data did not allow a reliable answer, the limitation is documented together with a recommended improvement.

Business questions

- Q1: Are prices really growing? (Year over Year)
- Q2: Is revenue growth driven by volume or price?
- Q3: Do we have an inventory problem?
- Q4: Does seasonality affect the margin?
- Q5: Do promotions have a negative impact on profit?
- Q6: Which countries are overstocked?
- Q7: Do we have suspicious data? (covered in the separate Data Quality Report)

Note on margin: margin / profit could not be reliably calculated, because of inconsistencies between `unit_price`, `base_price` and `discount_pct` (documented in the Data Quality Report). Where a question refers to margin or profit, sales volume or revenue is used as the closest reliable proxy.

Overview: Total Revenue per Year

Total revenue per year (the sum of `unit_price` * quantity) gives the big-picture trend before drilling into what drives it.

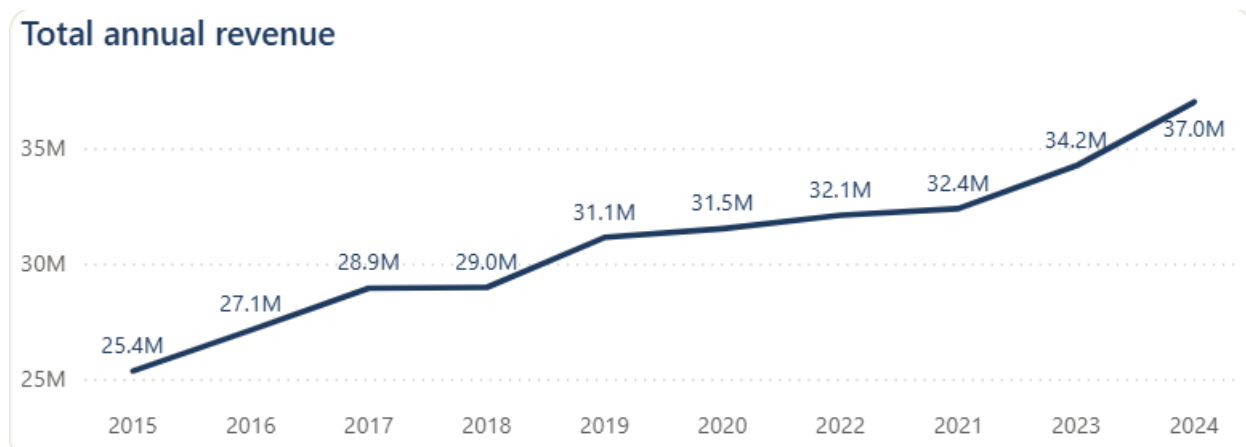


Figure 1. Total revenue, 2015–2024

Total annual revenue grows consistently across the whole period, from 25.4M in 2015 to 37.0M in 2024 an increase of about 46%. The next two questions explain what drives this growth: is it price or volume

Q1: Are prices really growing? (YoY)

Approach: average unit price per year, calculated from seasonality_VS_margin and grouped by YEAR(order_date). Unlike total revenue, the average price isolates price alone, independent of how many units were sold.

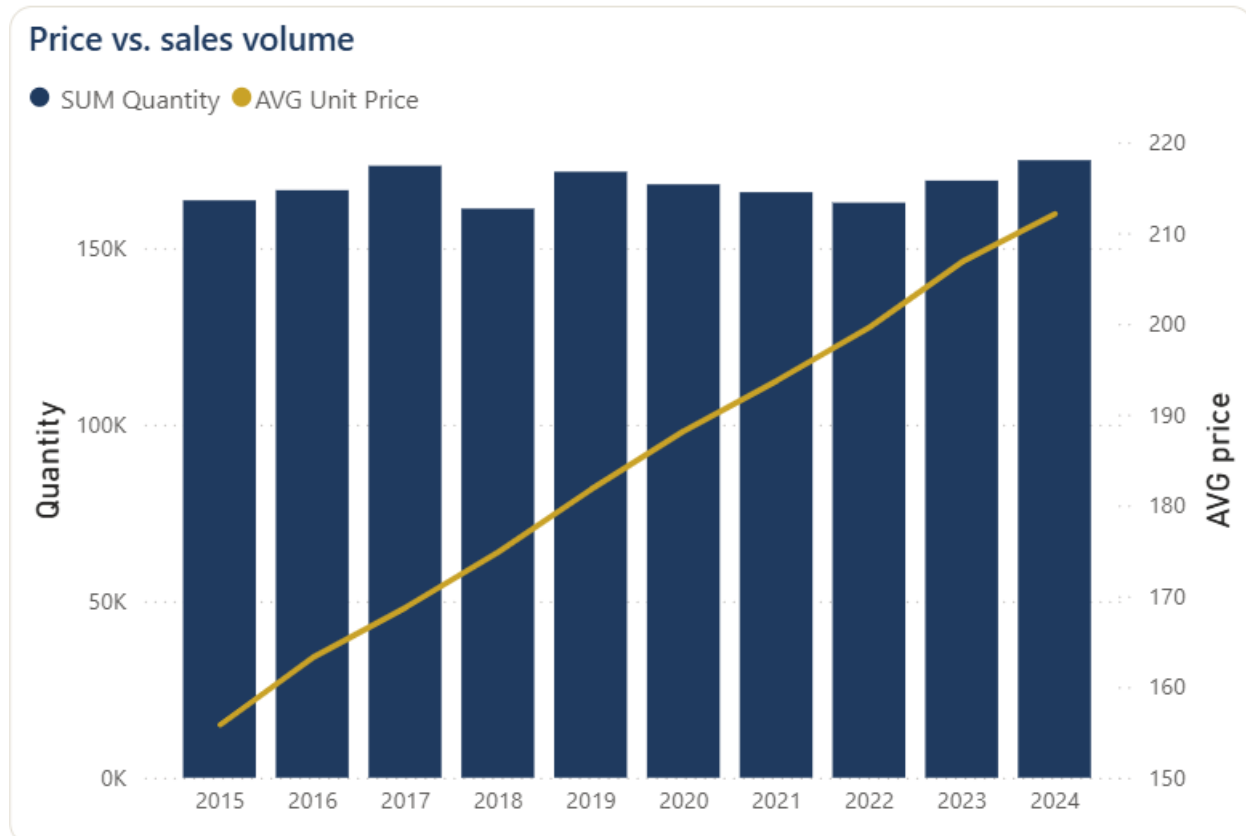


Figure 2. Comparison of average price to quantity of goods sold, 2015–2024

Finding: the average unit price rises steadily every year, from roughly 155 in 2015 to 212 in 2024, an increase of about 37% with no interruptions. Yes, prices are really growing.

Q2 : Is revenue growth driven by volume or price?

Compare yearly total quantity (bars) against average unit price (line) on one chart.

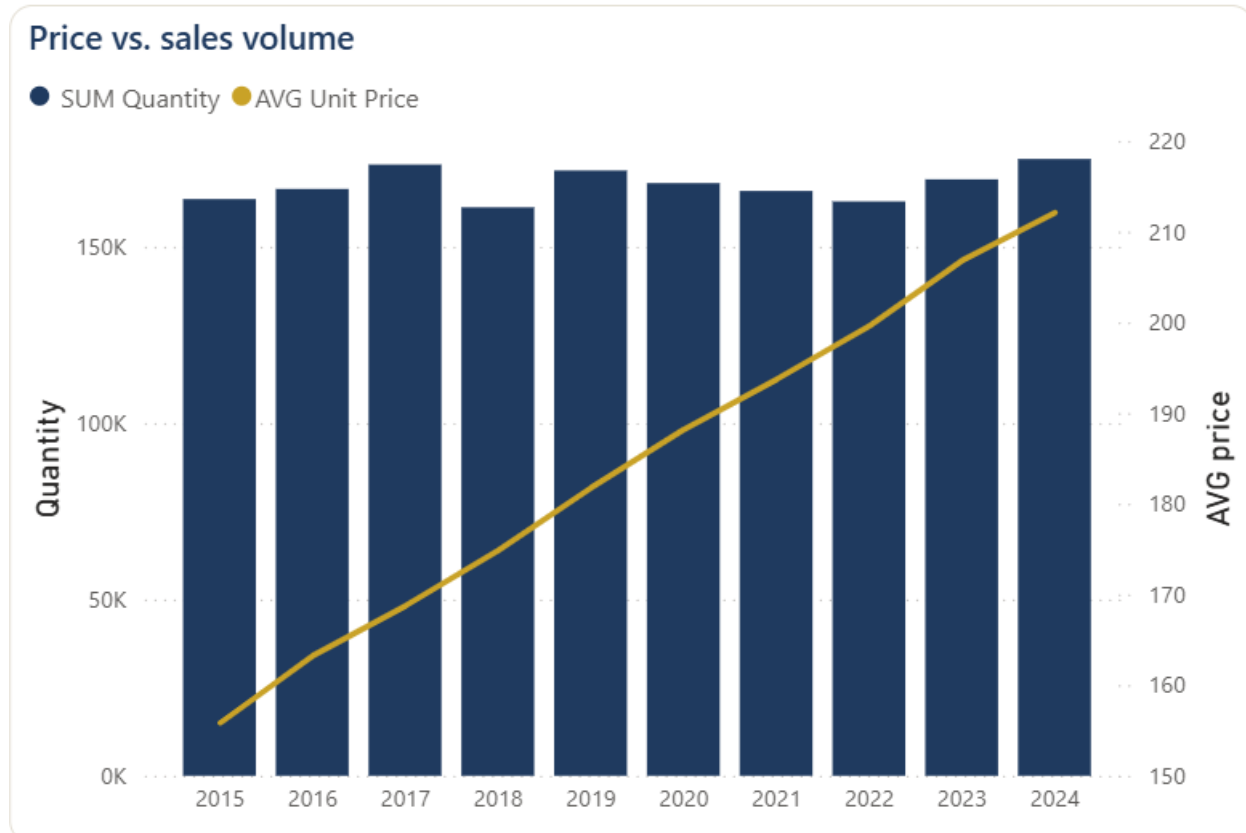


Figure 2. Comparison of average price to quantity of goods sold, 2015–2024

As can be seen on the chart the average price rises steadily (+37%), while total quantity stays flat, fluctuating only between about 161k and 174k units per year, with no upward trend. Revenue growth is therefore driven by rising prices, not by higher volume. This is clearest in 2018, where quantity dips below 2017 yet price keeps climbing confirming price, not volume, as the growth driver.

Q4: Does seasonality affect the margin?

Because margin cannot be reliably calculated (see note above), seasonality is analysed on sales volume. SUM of quantity by year, quarter and month.

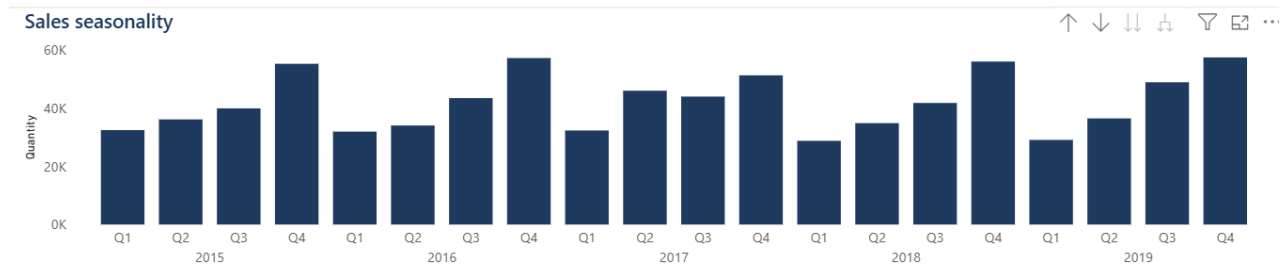


Figure 3. Sales seasonality by quarter, 2015–2019

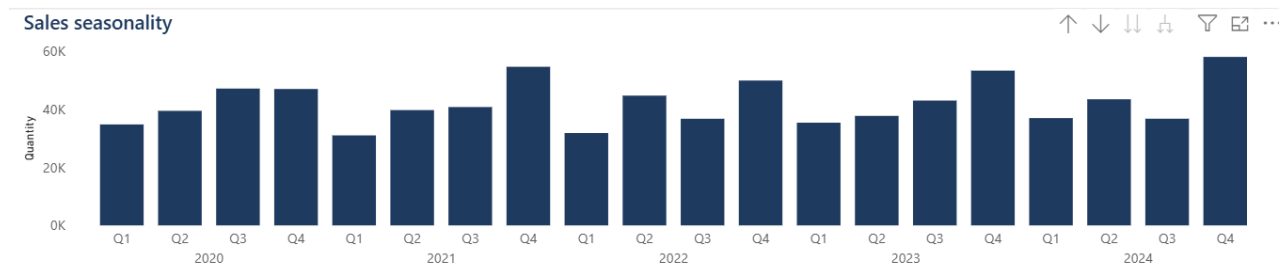


Figure 4. Sales seasonality by quarter, 2020–2024

Sales show a clear seasonal pattern. Q4 is the strongest quarter in almost every year (2015–2024); the only exception is 2020, where Q3 and Q4 are roughly equal. Within each year sales generally rise from Q1 toward Q4, though not strictly (in 2017, 2022 and 2024, Q2 exceeds Q3). The Q4 peak is most likely driven by Christmas demand, at month level, December is the peak and February the weakest month. The irregular shapes in 2022 and 2024 would be worth investigating further. Perhaps certain global events influenced the irregular shapes

Q5: Do promotions have a negative impact on profit?

Profit / margin cannot be reliably measured, so revenue is used as a proxy. Approach: monthly average discount versus monthly revenue, supported by a Pearson correlation coefficient.

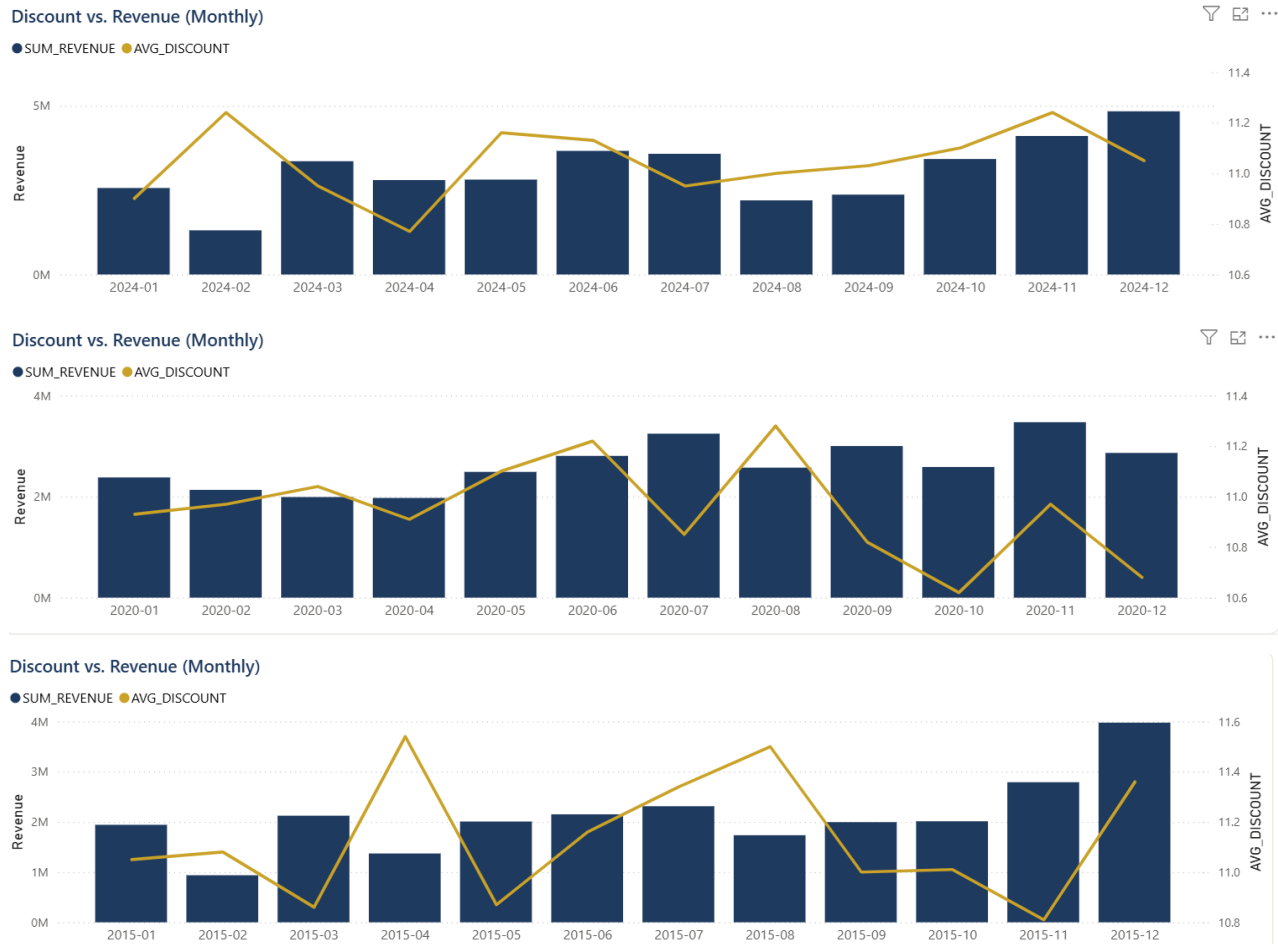


Figure 5. Discount vs. Revenue (2024,2020,2015)

The monthly average discount stays roughly flat (around 10–12%) while revenue varies strongly month to month, and the revenue peaks do not follow the discount line. The highest months are consistently Decembers, pointing to seasonality rather than promotions. To confirm this numerically, a Pearson correlation between monthly average discount and monthly revenue was calculated in SQL:

```
SELECT (AVG(disc * rev) - AVG(disc) * AVG(rev)) /  
       (STDEVP(disc) * STDEVP(rev)) AS correlation  
FROM (  
  SELECT CONCAT(YEAR(order_date), '-', MONTH(order_date)) AS ym,  
         AVG(discount_pct) AS disc, SUM(revenue) AS rev  
  FROM seasonality_VS_margin  
  GROUP BY YEAR(order_date), MONTH(order_date)  
) t;
```

	correlation
1	-0.012010474814912

The correlation is -0.01 effectively zero. There is no meaningful linear relationship between discount levels and revenue; discounting is not driving sales. (Caveat: Pearson correlation captures only linear relationships, but discount stays within a narrow 10-12% band, so this is not a concern here.)

Analysis Limitations: Q3 and Q6

Two questions could not be answered reliably with the current data. Instead of guessing here, I documented the limitation and paired with a recommended data improvement.

Q3: Do we have an inventory problem?

The inventory update date (last_stock_update) is not synchronised with the order date. This was quantified on the sales_inventory view. First, the share of rows where the two dates differ by more than a year:

```
SELECT COUNT(*) AS total_rows,
       SUM(CASE WHEN ABS(DATEDIFF(DAY, order_date, last_stock_update)) > 365
                    THEN 1 ELSE 0 END) AS rows_over_1_year,
       CAST(ROUND(100.0 * SUM(CASE WHEN ABS(DATEDIFF(DAY, order_date,
                    last_stock_update)) > 365 THEN 1 ELSE 0 END) / COUNT(*), 1)
            AS decimal(5,1)) AS pct_over_1_year
FROM sales_inventory;
```

	total_rows	rows_over_1_year	pct_over_1_year
1	378536	312923	82.7

Then the average and maximum size of the gap:

```
SELECT AVG(ABS(DATEDIFF(DAY, order_date, last_stock_update))) AS avg_gap_days,
       MAX(ABS(DATEDIFF(DAY, order_date, last_stock_update))) AS max_gap_days
FROM sales_inventory;
```

	avg_gap_days	max_gap_days
1	1542	3647

In 82.7% of joined rows (312,923 of 378,536) the inventory update date differs from the order date by more than a year. On average the gap is 1,542 days (4.2 years), reaching a maximum of 3,647 days (10 years). Stock levels therefore almost never reflect the situation at the time of sale, so no reliable conclusion about inventory health can be drawn. (Figures are at join-row level, because the view duplicates products across warehouses: see Q6.)

Q6: Which countries are overstocked?

The same product is stored in several warehouses, and orders are not linked to a specific warehouse. The scale of this was quantified. First, how many products sit in more than one warehouse:

```
SELECT COUNT(*) AS products_in_multiple_warehouses
FROM (
  SELECT product_id FROM inventory_mmmgkubv
  GROUP BY product_id
  HAVING COUNT(DISTINCT warehouse_country) > 1
) t;
```

	products_in_multiple_warehouses
1	1224

Then the total number of products, to express it as a share:

```
SELECT COUNT(DISTINCT product_id) AS total_products
FROM inventory_mmmgkubv;
```

	total_products
1	2491

Finding: 1,224 of 2,491 products (49%) are stored in more than one warehouse. Because orders are not linked to a fulfilling warehouse, joining sales with inventory duplicates each such order across every warehouse that holds the product. It is therefore impossible to tell which warehouse (country) a sale actually came from, and overstocked countries cannot be identified.

Recommended data improvement

The most impactful improvement would be to add a shipped_from_warehouse column to sales_orders, populated automatically when an order status changes to SHIPPED or COMPLETED. Adding this column would let us:

- Link each sale to the warehouse that actually fulfilled it
- Enable reliable inventory analysis (Q3) and overstocking detection by country (Q6)
- Support logistics optimisation and clearer, data-backed decisions for management

Summary and Recommendations

Key findings

- Revenue is growing steadily, driven by rising prices (average unit price +37% from 2015 to 2024), not by higher volume (quantity flat at 161k-174k units per year).
- Sales are strongly seasonal, peaking in Q4 (Christmas), with December the strongest and February the weakest month.
- Promotions show no measurable relationship with revenue (correlation -0.01); discounting is not driving sales.
- Inventory questions (Q3, Q6) cannot be answered reliably: stock-update dates are misaligned (82.7% off by more than a year, 4.2 years on average) and 49% of products sit in multiple warehouses with no order-to-warehouse link.

Recommendations for management

- Treat price, not volume, as the current growth lever and monitor whether continued price increases eventually start to suppress volume.
- Plan inventory, staffing and marketing around the strong Q4 / December peak.
- Reconsider the discount strategy. Current promotions do not appear to lift revenue.
- Add a shipped_from_warehouse column and synchronise stock-update timing to unlock reliable inventory and logistics analysis.